

How to Tax Capitalists in the Twenty-First Century?

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Capitalists in the Twenty-First Century

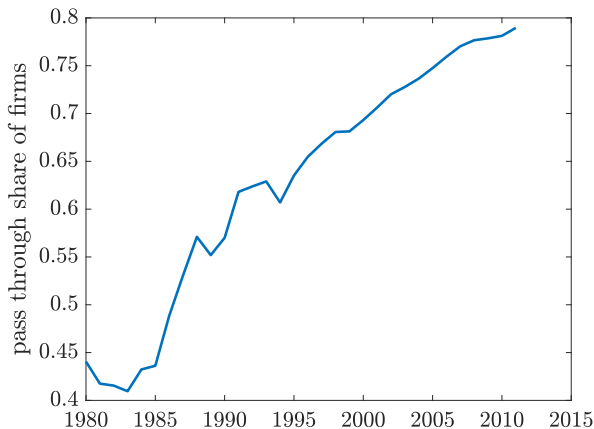
- **Secular shift** in the distribution of the legal forms of organization (LFO) of the U.S. businesses over the last 40 years.
- Profitable and closely held businesses have become dominant in the firms' population and their owners shape the right tail of income and wealth distribution.
- Predominantly organized as **pass-through businesses** - Dyrda, Pugsley (2019), Smith, Yagan, Zidar, Zwick (2019).
- Smith et. al. (2019) label them: **Capitalists in the Twenty-First Century**

Legal Forms of Organization (LFO) in the US

	Liability Protection	Ownership	Taxation of Profits
Sole Proprietorship	No	individual or family	Pass-through
General Partnership	No	general partners	Pass-through
Limited Partnership	No for partners Yes for limited part.	general and limited partners	Pass-through
Limited liability company	Yes	single or multiple members	Pass-through
S Corporation	Yes	one class of 1-100 domestic shareholders	Pass-through
C Corporation	Yes	no limit on number and type	Entity level

Key trade-off: tax and organizational simplicity versus flexibility to raise outside equity

Rise of the pass-throughs since 1980



Source: Authors calculations from Census LBD and Business Register

- Employment share of pass-throughs increased from **17.5** percent in 1980 to **65.4** percent in 2012. LBD Summary Statistics

The current U.S. Tax Code: key challenges

1. The code **distorts and insures** both labor and capital margins.
 - Personal income tax code **applies to both** labor income (wages and salaries) and capital income (interest income, dividends and some business profits).
 - Different **elasticities** and different **risks** associated with the two.
2. Business owners **choose their LFO** in response to the tax code modifications.
 - **Dyrda, Pugsley (2019)**: this margin is quantitatively relevant, flows were large following past reforms. Flows in the Past
3. **Multidimensional heterogeneity** which interacts with the tax code.
 - Workers and business owners are different "species" in the data. Differ among many dimensions.

This paper: How to Improve the Existing Code?

Design of the optimal tax system under two **revenue-neutral** scenarios:

1. Current legal framework
2. Uniform business profits taxation

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Design of the optimal tax system under two **revenue-neutral** scenarios:

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In a model featuring:

- **Endogenous choice of legal form** of business organization and selection across forms consistent with the data.
- Separation between **labor income risk** (productivity) and **capital income risk** (investment).
- Realistic representation of the current US tax code.
- **Heterogeneity** among workers and entrepreneurs.

Takeaway: unified, separate business profits tax

1. Optimal policy under **current legal restrictions**:

- The system is too restrictive to resolve conflicting interests of workers and entrepreneurs.
- Optimality calls for **elimination** of the corporate income taxes and **increase** of progressivity of the personal income tax code. Benefits workers via GE effects.

2. **Uniform Business Profits Tax**:

- **Welfare-dominant** over the current legal restrictions reform.
- **Separates** intertemporal distortion on capital accumulation from intratemporal distortions on labor supply.

Contribution to literature

1. Optimal taxation with workers: Domeij and Heathcote (2004), Conesa, Kitao, and Krueger (2009), Poschke et al. (2012), Krueger and Ludwig (2016), Heathcote et al. (2017a)

Contribution: Study the optimal policy problem in a model with workers and entrepreneurs.

2. Optimal taxation with entrepreneurs/firms: Panousi (2008), Meh and Terajima (2009), Panousi and Reis (2012), Evans (2014), Scheuer (2014)

Contribution: Endogenize the choice of the LFO in the optimal taxation problem.

3. LFOs in quantitative macro: Short and Glover (2019), Chen, Qi and Schlagenhauf (2018), Bhandari, McGratten (2021).

Contribution: Discipline the selection into the LFOs and study the optimal policy problem.

THE MODEL ECONOMY

Environment

- Unit measure of infinitely-lived households:
 - Fraction μ are workers.
 - Fraction $1 - \mu$ are entrepreneurs (Active Business Owners).
- Workers are subject to idiosyncratic labor productivity risk. Entrepreneurs are subject to idiosyncratic productivity risk. No aggregate risk.
- Incomplete markets with respect to idiosyncratic shocks.
- Entrepreneurs make endogenous choice of the legal form of organization.

Workers

Standard income fluctuation problem:

$$V^W(a, \varepsilon) = \max_{c, h, a'} u(c, 1 - h) + \beta \mathbb{E} [V^W(a', \varepsilon') | \varepsilon]$$

subject to

$$c + a' = a + y - T_y(wh\varepsilon) - \tau_d ra$$

$$y = ra + wh\varepsilon$$

$$a' \geq \underline{a}$$

a : savings

ε : stochastic labor productivity

$T_y(\cdot)$: income tax schedule

τ_d : dividend income tax

Stylized tradeoff between legal forms

C corporation:

Pro	Con
• Access to the supply of external equity • Completely diversified investment risk	• Profits subject to both corporate income and distribution taxes • Overhead costs

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Pass through:

Pro	Con
• Profits taxed once at personal income tax • Simple organization with no overhead costs	• Capital financed only through own equity • Undiversified investment risk

Entrepreneurs: technology and diversification

- DRS technology $f(k, n; z)$ homogeneous in k , n and z
- Gross profits:

$$\pi(z, k) = \max_n \{f(k, n; z) - wn\} = f_k k + f_z z$$

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C-corporation entrepreneur is fully diversified:

- Mutual fund chooses capital k^* at the end of the previous period before z was realized to equate

$$\mathbb{E}[(1 - \tau_c)(f_k(k^*; n^*; z) - \delta)] = r$$

- Entrepreneur receives preferred dividend

$$D(z, k^*) = (1 - \tau_c)(f_z(k^*; n^*; z)z - c_f)$$

where τ_c is the corporate income tax.

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Pass-through entrepreneur makes an investment decision and bears the idiosyncratic risk.

Entrepreneurs: C corporation (C)

Dynamic problem with pass through conversion option in continuation W^C :

$$\begin{aligned} V^C(a, k^*, z) &= \max_{s, c} u(c, 1 - \bar{h}) + \beta W^C(s, z) \\ &\text{subject to} \\ c + s &= a + y - \tau_d(ra + D(z, k^*)) \\ y &= ra + D(z, k^*) \\ s &\geq \underline{a} \end{aligned}$$

Dividend and risk free investment return taxed at τ_d

Income fluctuations from stochastic preferred dividend $D(z, k^*)$

Entrepreneurs: pass-through (P)

Dynamic problem with conversion option in continuation W^P

$$\begin{aligned} V^P(a, e, z) &= \max_{s, c} u(c, 1 - \bar{h}) + \beta W^P(s, z) \\ &\text{subject to} \\ c + s &= y + a + e - T_y(\pi - \delta e) - \tau_d r a \\ y &= r a + \pi(e, z) - \delta e \\ s &\geq \underline{a} \end{aligned}$$

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Homogeneity of technology in z , k and n implies:

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IFP from rents $f_z z$ and undiversified return on business equity $f_k e$

Continuation values: conversion and portfolio choice

Continuation value of the pass-through entrepreneur:

$$W^P(s, z) = \max \left\{ \mathbb{E}_C [V^C(s, k^*(z), z') | z] - f_s, \max_{e' \leq s - \bar{a}} \{ \mathbb{E}_P [V^P(s - e', e', z') | z] \} \right\}$$

Continuation value of the C-corp entrepreneur:

$$W^C(s, z) = \max \left\{ \mathbb{E}_C [V^C(s, k^*(z), z') | z], \max_{e' \leq s - \bar{a}} \{ \mathbb{E}_P [V^P(s - e', e', z') | z] - f_s \} \right\}$$

where f_s is a switching cost.

Aggregation and market clearings

- The number of pass-through owners $\mathbf{p}$ is determined by

$$\mathbf{p} = \mu \left(\int_{A \times E \times Z} d\lambda_P(a, e, z) \right)$$

and then the fraction of the C corporation owners is $(1 - \mu)(1 - p)$

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- Market clearing for labor requires

$$\begin{aligned} \int_A \int_{\epsilon} h(a, \epsilon) \epsilon d\lambda_w(a, \epsilon) &= \int_{A \times Z} n^*(z) d\lambda_C(a, z) \\ &+ \int_{A \times E \times Z} n(a, e, z) d\lambda_P(a, e, z) \end{aligned}$$

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- Market clearing for the capital stock requires (?) Foreign Holdings of Equity

$$\begin{aligned} \int_{A \times Z} k^*(z) d\lambda_C(a, z) + B &= \int_{A \times \epsilon} a'(a, \epsilon) d\lambda_w(a, \epsilon) + \int_{A \times Z} a'(a, z) d\lambda_C(a, z) \\ &+ \int_{A \times E \times Z} a'(a, e, z) d\lambda_P(a, e, z) \end{aligned}$$

TAKING THE MODEL TO THE DATA

Preferences and Technology

- Preferences:

$$u^i(c, h) = \frac{c^{1-\gamma}}{1-\gamma} - \mathbf{1}_{i=w} \psi \frac{h^{1+\frac{1}{\theta}}}{1+\frac{1}{\theta}} \quad i \in \{w, e\}$$

where $i = e$ is entrepreneur, $i = w$ is worker.

- Entrepreneurs, regardless of their legal form of business organization, have access to the DRS, production technology:

$$f(z, k, n) = z^{1-\nu} (k^\alpha n^{1-\alpha})^\nu$$

- Workers' productivity follows standard AR(1) process:

$$\log \varepsilon' = \mu_w + \rho_w \log \varepsilon + \eta_w$$

where $\eta_w \sim N(0, \sigma_w)$.

LFO-dependent productivity processes

The productivity Markov processes are:

$$\Gamma^P = \begin{bmatrix} p_{11} & \cdots & p_{1n} & 0 \\ \vdots & \ddots & \vdots & 0 \\ p_{n1} & \cdots & p_{nn} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \Gamma^C = \begin{bmatrix} p_{11}(1-f_1) & \cdots & p_{1n}(1-f_1) & f_1 \\ \vdots & \ddots & \vdots & \vdots \\ p_{n1}(1-f_n) & \cdots & p_{nn}(1-f_n) & f_n \\ 0 & 0 & 1-q & q \end{bmatrix}$$

where

- $p_{i,j}$ for $i, j = 1, \dots, n$ are elements of the matrix obtained through the discretization procedure.

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- $q \in [0, 1]$ is the probability of staying in the top state for the C corporation
- probability of switching to the top state for the C corporation are given by

$$f_i = \theta_{f1} \left(\frac{z_i}{z_n} \right)^{\theta_{f2}} \quad i \in \{1, \dots, n\}$$

- we allow for $q = 0$ and $\theta_{f1} = 0$ in calibration which makes $\Gamma^P = \Gamma^C$.

Switching costs: extreme value shocks

- Switching costs: i.i.d. with a logistic distribution with mean f_s and dispersion parameter σ_f .
- Continuation value of the pass-through entrepreneur is then:

$$W^P(s, z) = \sigma_f \ln \left\{ \exp \left\{ \frac{\mathbb{E}_C [V^C(s, k^*(z), z') | z] - f_s}{\sigma_f} \right\} + \exp \left\{ \frac{\max_{e' \leq s - \bar{a}} \mathbb{E}_P [V^P(s - e', e', z') | z]}{\sigma_f} \right\} \right\}.$$

- Decision rule becomes conditional choice probability:

$$\Pr(C | s, P) = \frac{\exp \left\{ \frac{\mathbb{E}_C [V^C(s, k^*(z), z') | z] - f_s - \max_{e' \leq s - \bar{a}} \mathbb{E}_P [V^P(s - e', e', z') | z]}{\sigma_f} \right\}}{1 + \exp \left\{ \frac{\mathbb{E}_C [V^C(s, k^*(z), z') | z] - f_s - \max_{e' \leq s - \bar{a}} \mathbb{E}_P [V^P(s - e', e', z') | z]}{\sigma_f} \right\}}$$

and $W^C(s, z)$, $\Pr(C | s, P)$ are determined symmetrically.

Tax system in the initial equilibrium

- **Income tax schedule** from Heathcote-Storesletten-Violante:

$$T(y) = y - \lambda_y y^{1-\tau_y}$$

where:

- τ_y controls progressivity of the code. We estimate $\tau_y = 0.097$.
- λ_y is set to balances the budget so that Gov. Revenues/GDP is 21%.
- **Dividend income tax:** $\tau_d = 26.3\%$ to match average marginal tax rate on dividends in TAXIM.
- **Corporate income tax:** $\tau_d = 19.7\%$ to match average effective tax rate estimated from NIPA.
- **Debt to GDP:** 102% to match the 2013-2017 average

Model Fit

Statistic	Source	Model	Target
Capital/Output*	NIPA	1.25	1.25
Avg Labor Supply	CPS	0.34	0.33
Gini Income	SCF	0.62	0.65
Gini Ent Income	SCF	0.64	0.67
Gini Wor Income	SCF	0.59	0.62
Top 1 Ent Inc Share	SCF	0.20	0.27
Top 10 Ent Inc Share	SCF	0.49	0.53
Fraction of P ent	LBD	0.80	0.81
Emp Share of P ent	LBD	0.64	0.57
Flow CP	LBD	0.02	0.01
Flow PC	LBD	0.004	0.002
Logit CH	SCF	-0.94	-1.08
Logit Prof.	SCF	-0.82	-0.93
Logit CH&Prof.	SCF	0.17	0.06

* We define the capital stock in the data as the sum of private fixed assets and durable consumption.

Selection: who is who via indirect inference

- Split the SCF population into workers and Active Business Owners (ABO) and attach the legal form of organization to each ABO.
- Run for the ABOs the logistic regression:

$$\Pr(D_{it} = 1) = F(\mu_t + \gamma_1 \log \Pi_{it} + \gamma_2 \log X_{it} + \gamma_3 \log \Pi_{it} \times \log X_{it}) \quad (1)$$

where

D_{it} : pass-through dummy

μ_t : year fixed effect

Π_{it} : profits

X_{it} : net worth

$$F(x) = \frac{e^x}{1+e^x}$$

$t \in \{2004, 2007, 2010, 2013, 2016\}$: SCF waves

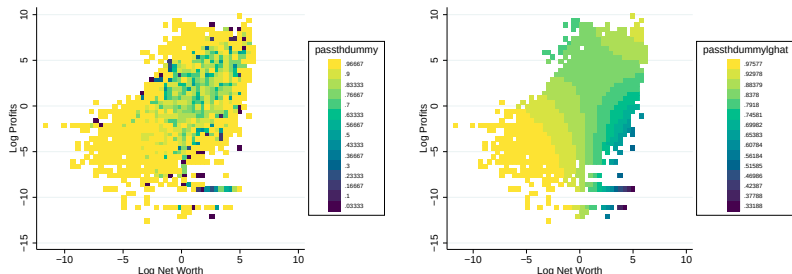
Selection: model vis a vis the data.

	Data		Model
	(1) <i>passth_{it}</i>	(2) <i>passth_{it}</i>	
$\log \text{networth}_{it}$	-0.955 (0.183)	-0.927 (0.184)	-0.823
$\log \text{profit}_{it}$	-1.095 (0.132)	-1.081 (0.133)	-0.944
$\log \text{networth}_{it} \times \log \text{profit}_{it}$	0.064 (0.012)	0.062 (0.012)	0.170
N	27,507	27,507	
R^2	0.062	0.065	
Time FE	No	Yes	

Note: Pooled SCF waves 2004-2016; estimated parameters of logistic regression; robust standard errors in parentheses; R^2 measure is pseudo- R^2 .

Selection: who is who in the SCF?

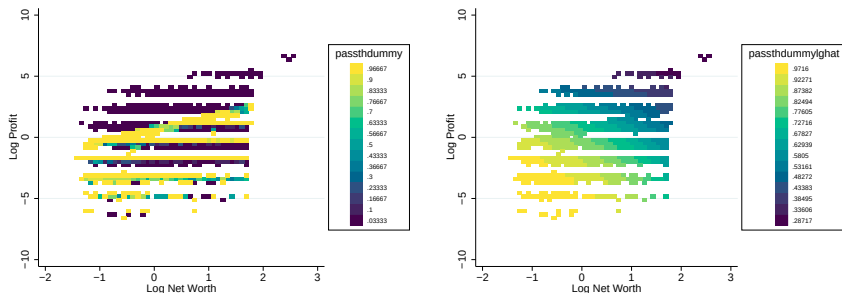
Figure: Conditional Probability of observing the pass-through - empirical distribution (left panel), logit regression (right panel)



Notes: SCF waves 2004-2019, the variables are deviations from annual average

Selection: who is who in the model?

Figure: Conditional Probability of observing the pass-through - empirical distribution (left panel), logit regression (right panel)



Notes: The variables are deviations from average

THE OPTIMAL POLICY

Social Welfare Function

- Workers

$$SWF_w(\mathcal{T}) = \int_{A \times E} V_1^W(a, \varepsilon; \mathcal{T}) d\lambda_0(a, \varepsilon)$$

where $V_1^W(a, \varepsilon; \mathcal{T})$ is the value function in the first period of the transition induced by new tax system $\mathcal{T}$ and $\lambda_0(a, \varepsilon)$

- Entrepreneurs

$$SWF_e(\mathcal{T}) = \int_{X \times Z} p V_1^P(x, z; \mathcal{T}) d\lambda_{0P}(x, z) + (1 - p) V_1^C(x, z; \mathcal{T}) d\lambda_{0C}(x, z)$$

- Population welfare

$$SWF(\mathcal{T}) = \mu SWF_w(\mathcal{T}) + (1 - \mu) SWF_e(\mathcal{T})$$

where $\mu=0.88$ is the fraction of workers in the population.

Optimal Tax System

- The optimal tax reform is the sequence $\mathcal{T}^* = \{\tau_{c,t}, \tau_{d,t}, \tau_{y,t}, \lambda_{y,t}\}_{t=0}^{\infty}$ that solves:

$$\mathcal{T}^* \in \arg \max_{\mathcal{T} \in \Gamma} SWF(\mathcal{T}) \quad (2)$$

- Solving (2) is a bit challenging (Dyrda, Pedroni (2022) for SIM model). More pragmatic approach: one-time policy change, i.e.:

$$\tau_{c,t} = \tau_{c,1}, \tau_{y,t} = \tau_{y,1}$$

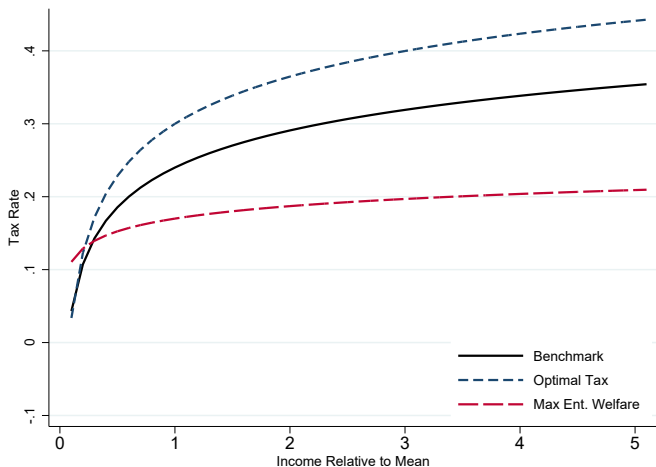
- Adjust $\lambda_{y,t}$ so that the reform scenarios are **revenue neutral**. Keep dividend income tax $\tau_{d,t}$ and interest rate fixed.
- Today** a limited version: max welfare in stationary equilibrium, but less relevant in SOE.

Current Framework: the optimal tax schedules

	Baseline Economy	Optimal Tax System	Max. Entr. Welfare
Progressivity, τ_y	0.10	0.14	0.03
Corporate Income Tax, τ_c	0.20	0.00	0.00
Fiscal closure, $1 - \lambda_y$	0.24	0.30	0.17
Debt to GDP	1.02	1.02	1.02
Revenues to GDP	0.21	0.21	0.21
$\Delta SWF_w(\%)$	-	1.37	-1.09
$\Delta SWF_c(\%)$	-	7.01	8.84
$\Delta SWF(\%)$	-	2.05	0.12
% of Pass-Throughs	80.4	14.0	28.1

- **Trade-off:** distortions on capital accumulation vs. distortions of labor and redistribution/insurance provision for workers
- **Increase of progressivity + elimination of τ_c** benefits both workers and entrepreneurs

Current Framework: the optimal tax schedules



Current Framework: macro aggregates (% changes)

	Baseline Economy	Optimal Tax System	Max. Entr. Welfare
Employment	0.0	-0.7	1.7
Output	0.0	3.3	4.9
Capital	0.0	17.7	17.9
Wage	0.0	4.0	3.1
Employment C	0.0	271.0	223.2
Employment P	0.0	-88.6	-70.0
Output C	0.0	285.8	233.3
Output P	0.0	-88.2	-69.0
% of Pass-Throughs	80.4	14.0	28.1

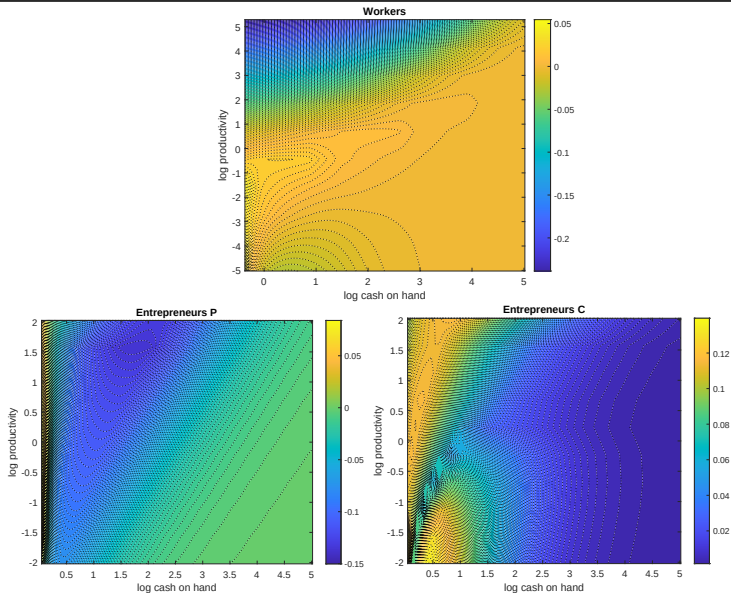
- Optimal Tax System: workers benefit higher wages, allocation of capital improved.
- Value added and employment reallocated towards C corporations.

Inequality Implications

	Baseline Economy	Optimal Tax System	Max. Entr. Welfare
Gini Population	0.62	0.58	0.60
Top 1% Share (%)	15.2	16.3	16.2
Top 10% Share (%)	45.1	43.5	44.0
Gini Workers	0.59	0.60	0.61
Gini Entrepreneurs	0.64	0.71	0.78
% of ABOs in Top 10%	40.0	25.6	33.1
% of Pass-Throughs	80.4	14.0	28.1

- Income dispersion **within** entrepreneurs increases.
- Wage boost makes workers and entrepreneurs **more similar** in terms of income.

Conditional welfare: who gains, who loses?



Uniform Business Profit Tax in the Model

- Abandon double-taxation of profits.
- Introduce the same-flat tax rate τ_b on **business profits**.
- What changes in the model?

$$\text{C ent: } c + s = (1 + r)a - \tau_d(ra + (1 - \tau_c)(f_z(k^*; n^*; z')z' - c_f))$$

$$\text{P ent: } c + s = \pi + (1 - \delta)e + (1 + r)a - \tau_dra - T_y(\pi - \delta e)$$

- The number of instruments **remains unchanged** (2 instruments set optimally). Just redefining the tax base.
- Again **budget-neutral** reform.

Uniform Business Profit Tax in the Model

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$$\text{C ent: } c + s = (1 + r)a - \tau_d ra + (1 - \tau_b)(f_z(k^*; n^*; z')z' - c_f)$$

$$\text{P ent: } c + s = e + (1 + r)a - \tau_d ra + (1 - \tau_b)(\pi - \delta e)$$

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- Again **budget-neutral** reform.

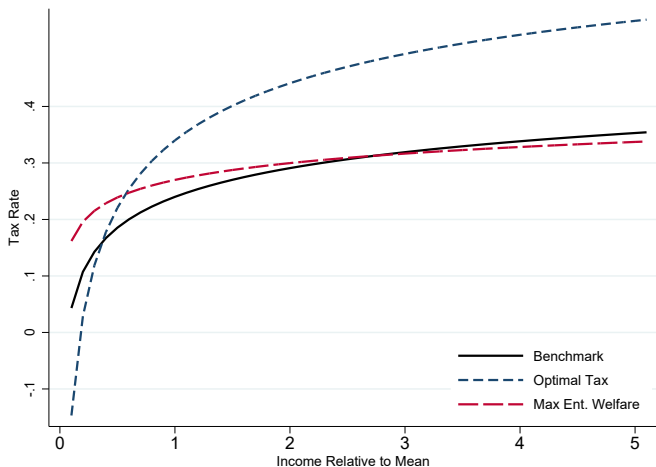
Uniform Business Tax: the optimal tax schedules

	Baseline Economy	Optimal Tax System	Max. Entr. Welfare
Progressivity, τ_y	0.10	0.24	0.06
Uniform Business Tax, τ_b	—	0.26*	0.09
Fiscal closure, $1 - \lambda_y$	0.24	0.34	0.27
Debt to GDP	1.02	1.02	1.02
Revenues to GDP	0.21	0.21	0.21
$\Delta SWF_w(\%)$	-	2.63	-4.38
$\Delta SWF_e(\%)$	-	12.72	29.11
$\Delta SWF(\%)$	-	3.85	-0.32

* Should be compared with 0.20 is a corporate income tax + 0.26 of dividend tax.

- Separation of the codes **welfare-dominates** the existing code. Both workers and entrepreneurs gain.
- More insurance and redistribution among workers via sharp **increase in progressivity**. Overall tax burden **lower** for large entrepreneurs.

Uniform Business Tax: the optimal tax schedules



Uniform Business Tax: macro aggregates (% changes)

	Baseline Economy	Optimal Tax System	Max. Entr. Welfare
Employment	0.0	-3.9	2.2
Output	0.0	-1.2	4.6
Capital	0.0	5.2	14.5
Wage	0.0	2.8	2.3
Employment C	0.0	-80.0	134.0
Employment P	0.0	20.7	-40.4
Output C	0.0	-79.5	139.4
Output P	0.0	24.1	-39.0
% of Pass-Throughs	80.4	91.0	48.7

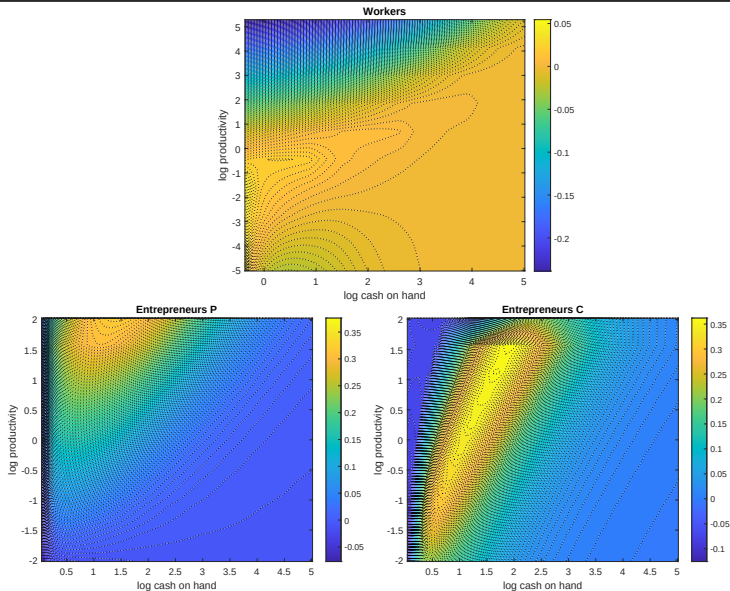
- Output and employment **fall** following the reform (high distortions on lab supply), reversed for Max. Ent. welfare.
- Reallocation of value added towards pass-through businesses, reversed for Max. Ent. welfare.

Inequality Implications

	Baseline Economy	Optimal Tax System	Max. Entr. Welfare
Gini Population	0.62	0.63	0.60
Top 1% Share (%)	15.2	16.3	16.9
Top 10% Share (%)	45.1	47.3	44.8
Gini Workers	0.59	0.57	0.61
Gini Entrepreneurs	0.64	0.62	0.73
% of ABOs in Top 10%	40.0	42.5	38.2
% of Pass-Throughs	80.4	95.0	48.7

- Reform increases inequality by moving **more entrepreneurs** towards the top of the income distribution.

Conditional welfare: who gains, who loses?



Conclusions

- Study the design of the optimal tax system taking seriously the nature of business and labor income and the **endogenous choice of legal form of business organization**.
- We find existing tax code can not resolve the tensions between workers and entrepreneurs.
- **Key policy prescription:** abandon double taxation of profits and separate taxation of labor income and business profits (labor distortion and labor risk vs. capital distortions + investment risk).

ADDITIONAL SLIDES

LBD Summary Statistics

	1980-1984	1985-1989	1990-1994	1995-1999	2000-2004	2005-2009
<i>Average size (employees)</i>						
C corporations	23.12	18.25	19.62	19.68	19.83	19.06
S corporations	10.67	13.94	13.91	13.17	12.63	11.99
Partnerships	8.44	9.33	11.34	12.53	17.14	18.35
Sole proprietors	3.94	4.07	4.14	4.37	4.89	5.46
<i>Exit rate (percent)</i>						
C corporations	11.11	9.97	8.68	8.56	9.03	9.27
S corporations	14.51	10.83	8.71	8.67	8.57	9.42
Partnerships	22.20	19.67	16.18	15.99	14.35	14.23
Sole proprietors	20.22	17.26	15.55	16.35	16.10	17.44
<i>Share of employers (percent)</i>						
C corporations	55.59	50.05	39.52	34.83	29.27	24.15
S corporations	9.27	15.77	26.35	33.35	39.80	45.44
Partnerships	7.78	7.90	6.70	6.91	9.61	12.64
Sole proprietors	27.36	26.27	27.42	24.91	21.32	17.78

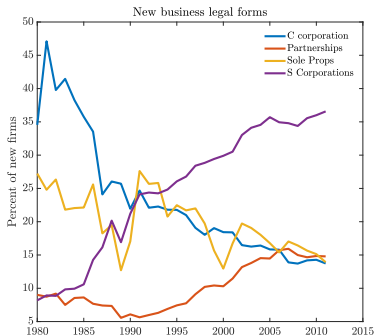
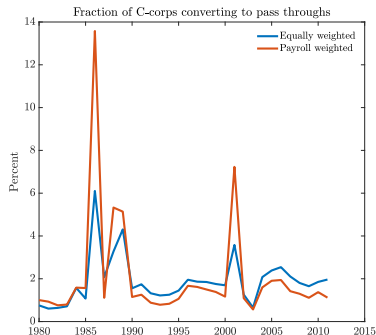
Model Parametrization

Parameter	Symbol	Discipline	Value
Externally calibrated			
Fraction of workers	μ	SCF data	0.88
Risk aversion	γ	-	1.50
Frisch elasticity	θ	-	0.85
Depreciation rate	δ	NIPA	0.08
Interest rate	r	Jorda et. al (2019)	0.02
Internally calibrated			
Discount factor	β	Targets in Table	0.98
Returns to scale	ν	Targets in Table	0.82
Persistence ent.	ρ_z	Targets in Table	0.96
Std ent. product.	σ_z	Targets in Table	0.44
Mean prod. wor.	μ_w	Targets in Table	2.59
Persistence wor.	ρ_w	Targets in Table	0.95
Std wor. product.	σ_w	Targets in Table	0.29
Fixed cost C corp.	c_f	Targets in Table	0.15
Mean switching cost	f_s	Targets in Table	27.14
Extreme value shock std	σ_s	Targets in Table	5.18
Scaling C corp. productivity	θ_z	Targets in Table	1.09
Probability shifter for C corp.	θ_{f1}	Targets in Table	0.29
Probability power for C corp.	θ_{f2}	Targets in Table	18.61
Probability of staying in the top state	q	Targets in Table	0.35

LBD - estimating firm level transitions

1. US Census Bureau Longitudinal Business Database (LBD) and linked Business Register (BR)
 - Near universal coverage of the nonfarm private sector
 - Longitudinally linked at the establishment level and aggregated to firms
 - Linkages robust to changes in ownership and LFO
2. Using LBD and linked BR record 4 possible legal forms: **C corporation, Partnerships (General/LLC/LLP), Sole Proprietors, and S corporation.**
3. Estimate transition matrix across these states plus an entry/exit state for the years 1980 to 2012 using empirical distribution.

Increases in pass throughs around major tax reforms



Source: Census LBD and Business Register

- Conversions surge around major tax reforms: Tax Reform Act of 1986, Economic Growth and Tax Relief Reconciliation 2001.
- Both reduced personal income tax rates, relative to the dividend and corporate income tax.

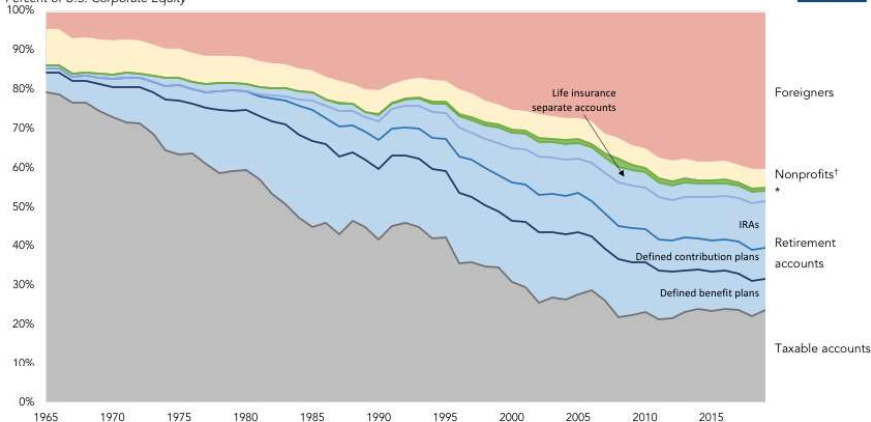
Who is left to tax?

[Back](#)

Figure 1: Ownership of U.S. Corporate Stock, 1965-2019

Direct and Indirect Holdings

Percent of U.S. Corporate Equity



Source: Board of Governors of the Federal Reserve System, "Financial Accounts of the United States;" Tax Policy Center Calculations

† Dashed lines indicate TPC estimates

* Federal, state and local government holdings, including equity in 529 college savings plans



Government

- Tax Revenues:

$$\begin{aligned}R_i &= \int_{A \times \epsilon} T_i(wh\epsilon) \, d\lambda_w(a, \epsilon) + \int_{A \times E \times Z} T_i(\pi(e, z) - \delta e) \, d\lambda_P(a, e, z) \\R_d &= \int_{A \times Z} T_i(D(z) + ra) \, d\lambda_C(a, z) + \int_{A \times E \times Z} T_i(ra) \, d\lambda_P(a, e, z) \\&\quad + \int_{A \times \epsilon} T_i(ra) \, d\lambda_w(a, \epsilon) \\R_c &= \int_{A \times Z} \tau^c(\pi(k^*(z); z) - c_f) \, d\lambda_C(a, z)\end{aligned}$$

- Government budget constraint:

$$G + (1 + r) B = B' + R_i + R_c \tag{1}$$

Portfolio choice: private equity expected return

Pass-through allocates savings s to solve

$$\max_{e' \leq s - \bar{a}} \{ \mathbb{E} [V^P (s - e', e', z')] \}$$

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Choose e' so after-tax net expected return on private equity

$$\mathbb{E} [(1 - T'_y) (f_k - \delta) | z] = (1 - \tau_d) r - \frac{\text{Cov} [u_c, (1 - T'_y) f_k | z]}{\mathbb{E} [u_c | z]} + \frac{\xi}{\beta \mathbb{E} [u_c | z]}$$

Multiplier ξ on capital constraint $\xi(s - \bar{a} - e') = 0$

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